

Probability and Statistics

Course Code: STAT-2311

Chapter: 5

Definition:

1. Experiment

- Experiment is an act that can be repeated under given conditions. Tossing of a coin or throwing of a dice or the drawing of a cards etc, are the example of experiment.

2. Random experiment

- Experiments are called random experiments if the outcomes depend on chance and cannot be predicted with certainty.
Example: Tossing of a fair coin, throwing of dice etc are the examples of random experiments.

3. Sample space

- The collection or totality of all possible outcomes of a random experiment is called sample space. It is usually denoted by S or Ω .
If we toss a coin, the sample space is $S = \{H, T\}$ where H and T denote the head and tail of the coin respectively.
If we toss a coin two times, then the sample space is $S = \{HH, HT, TH, TT\}$.

4. Sample point

- Each element of a sample space is called sample point.

5. Event

- Any subset of a sample space is called event. There are two types of event: Simple event and Compound event

6. Mutually exclusive event

- Two events are said to be mutually exclusive if they have no common points. If A and B are two mutually exclusive events, then $AB = \emptyset$.

7. Non- mutually exclusive event

- Two events are said to be not mutually exclusive event if they have common points. If A and B are two not mutually exclusive events, then $AB \neq \emptyset$.

8. Conditional Probability

- If A and B are two events in S . Then the conditional probability of A for given value of B , denoted by $P[A | B]$ is defined by

$$P[A | B] = \frac{P[AB]}{P[B]} ; \quad P[B] > 0$$

$$P[B | A] = \frac{P[AB]}{P[A]} ; \quad P[A] > 0$$

9. Independent Event

➤ Two events A and B are said to be independent if and only if one of the following conditions holds:

- a) $P[AB] = P[A] P[B]$
- b) $P[A | B] = P[A]$
- c) $P[B | A] = P[B]$

10. Dependent Event

➤ Two events A and B are said to be dependent if and only if one of the following conditions holds:

- a) $P[AB] \neq P[A] P[B]$
- b) $P[A | B] \neq P[A]$
- c) $P[B | A] \neq P[B]$

Definition of probability

There are four approaches of defining probability.

1. Classical or mathematical or priori definition of probability.

$$P(A) = \frac{\text{Favorable number of outcome}}{\text{Total number of outcome}}$$

$$= \frac{N(A)}{N(S)} = \frac{m}{n}; \quad 0 \leq P(A) \leq 1$$

If A be the event in S then the probability of the event A denoted by P[A] is defined as

$$P(A) = \frac{N(A)}{N(S)} = \frac{m}{n}$$

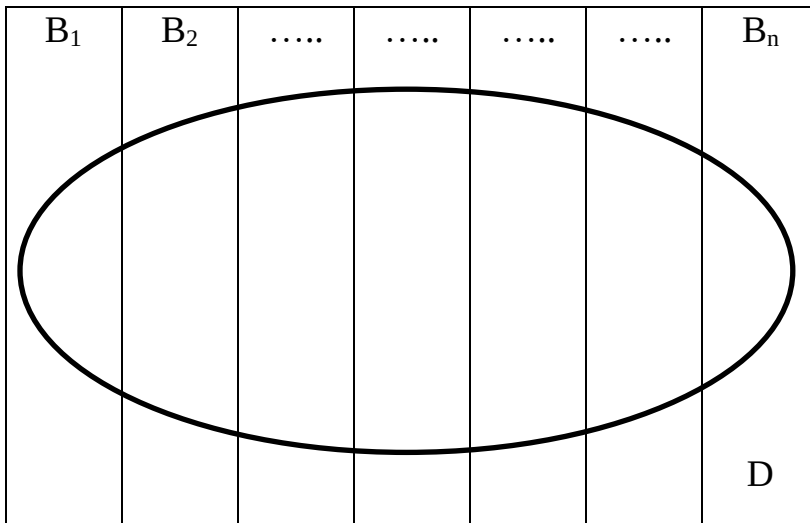
2. Axiomatic probability

- a) $P(A) \geq 0$
- b) $P(S) = 1$
- c) $P[A \cup B] = P[A] + P[B]$

Statement of Bayes Theorem

Let $B_1, B_2, \dots, B_n$ be n mutually exclusive and exhaustive events in a random experiment and D be any event in S, then Bayes theorem state that,

$$P[B_i|D] = \frac{P[B_i]P[D/B_i]}{\sum_{j=1}^n P[B_j]P[D/B_j]}; i = 1, 2, \dots, n$$



Math Problem:

In a factory, machines A, B and C manufacture respectively 25%, 35% and 40% of the product. Of the total of their output 5, 4 and 2 percent are defective bolts respectively. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by machine (i) A and (ii) C.

Solution:

Let

B_1 : An item produced by machine A

B_2 : An item produced by machine B

B_3 : An item produced by machine C

D : Defective item produced by the machines.

We have,

$$P[B_1] = 25\% = 0.25,$$

$$P[B_2] = 35\% = 0.35,$$

$$P[B_3] = 40\% = 0.40,$$

And,

$$P(D|B_1) = 5\% = 0.05$$

$$P(D|B_2) = 4\% = 0.04$$

$$P(D|B_3) = 2\% = 0.02$$

We have to find,

$$P(B_1|D) \text{ and } P(B_3|D)$$

$$P(B_1|D) = \frac{P(B_1)P(D|B_1)}{P(B_1)P(D|B_1) + P(B_2)P(D|B_2) + P(B_3)P(D|B_3)}$$

$$\begin{aligned}
 &= \frac{(0.25)(0.05)}{(0.25)(0.05) + (0.35)(0.04) + (0.40)(0.02)} \\
 &= 0.3623
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 P(B_3|D) &= \frac{P(B_3)P(D|B_3)}{P(B_1)P(D|B_1) + P(B_2)P(D|B_2) + P(B_3)P(D|B_3)} \\
 &= \frac{(0.40)(0.02)}{(0.25)(0.05) + (0.35)(0.04) + (0.40)(0.02)} \\
 &= 0.2319
 \end{aligned}$$